

Model-agnostic variable importance for survival outcomes: a simulation study under additive and mixed hazards

Simulation protocol

Stacy Cyrille

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1 Introduction

We conducted a simulation study to examine the behavior of Wolock’s model-agnostic exclusion variable importance measure (VIM) under a range of survival data-generating mechanisms. The objectives were to evaluate its performance and inferential validity under additive hazards, mixed additive–multiplicative hazards, and proportional hazards settings, including high-dimensional scenarios with rare binary covariates and heterogeneous effect sizes. We also compared alternative strategies for nuisance function estimation and contrasted exclusion-based importance with permutation-based importance across all scenarios.

2 Objectives

The simulation study was designed to evaluate the performance, robustness, and inferential validity of Wolock’s model-agnostic exclusion variable importance measure (VIM) across several survival data-generating mechanisms.

First, under an **additive hazards framework**, we assess the accuracy and inferential properties of the VIM and compare different strategies for estimating the conditional survival functions, namely *global survival stacking*, *survival Super Learner*, the *Aalen additive hazards model*, and *random survival forests*.

Second, under **mixed additive and multiplicative hazard structures**, we investigate the behavior of the VIM when both types of effects coexist. In this setting, conditional survival functions are estimated using *global survival stacking*, *survival Super Learner*, the *Cox–Aalen model*, and *random survival forests*.

Third, we assess the sensitivity of the VIM when covariates correspond to **rare alterations** with **heterogeneous effect sizes**, ranging from weak to strong, under a **proportional hazards model**.

3 Data-generation mechanisms

3.1 Simulation of covariates

We simulated datasets combining binary and continuous covariates for $i = 1, \dots, n$ individuals.

3.1.1 Binary covariates

Binary covariates were generated as correlated Bernoulli variables:

$$X_{ij} \in \{0, 1\}, \quad j = 1, \dots, p_B,$$

where $X_{ij} = 1$ indicates the presence of binary feature j .

Each variable had marginal prevalence

$$\pi_j = \Pr(X_{ij} = 1).$$

To induce correlation between variables, a latent Gaussian copula model was used:

$$\mathbf{Z}_i \sim \mathcal{N}(0, \Sigma_B),$$

where Σ_B is a block-structured correlation matrix.

Binary variables were obtained through probit thresholding:

$$X_{ij} = \mathbb{I}(Z_{ij} < \Phi^{-1}(\pi_j)),$$

where Φ^{-1} denotes the quantile function of the standard normal distribution. This construction guarantees

$$\Pr(X_{ij} = 1) = \pi_j.$$

Thus, marginal prevalences are exactly controlled while inducing dependence between binary variables through the latent correlation matrix.

3.1.2 Continuous covariates

Continuous covariates were simulated from a multivariate normal distribution:

$$\mathbf{W}_i \sim \mathcal{N}(0, \Sigma_C),$$

where $\Sigma_C \in \mathbb{R}^{p_C \times p_C}$ is a block-correlation matrix.

Variables were partitioned into K blocks of equal size b . For block k , define

$$B_k = \{(k-1)b + 1, \dots, kb\}.$$

The correlation matrix $\Sigma_C = (\sigma_{jl})$ was defined as

$$\sigma_{jl} = \begin{cases} 1 & \text{if } j = l, \\ \rho_k & \text{if } j \neq l \text{ and } j, l \in B_k, \\ \rho_B & \text{if } j \in B_k, l \in B_m, k \neq m. \end{cases}$$

Equivalently, each diagonal block can be written as

$$\Sigma_{C,k} = (1 - \rho_k)I_b + \rho_k \mathbf{1}_b \mathbf{1}_b^\top,$$

where I_b denotes the $b \times b$ identity matrix and $\mathbf{1}_b$ is a vector of ones of length b .

Continuous covariates were therefore correlated according to a block dependence structure.

3.2 Simulation of time-to-event outcomes

Let T denote the event time and C an independent censoring time. The observed data consisted of (Y, Δ, X, Z) , where:

$$Y = \min(T, C), \quad \Delta = \mathbb{I}(T \leq C).$$

Censoring times were generated independently from an exponential distribution to achieve a target censoring proportion.

Additive Aalen model

Time-to-event outcomes were generated under an additive hazards framework.

For each individual $i = 1, \dots, n$, with covariate vector $\mathbf{X}_i$, the hazard function was defined as

$$\lambda(t | X_i) = \lambda_0(t) + X_i^\top \alpha,$$

where $\lambda_0(t)$ denotes the baseline hazard function and α represents the vector of additive hazard effects.

Cox-Aalen model

To evaluate performance under mixed hazard structures, we additionally generated data from a Cox-Aalen model:

$$\lambda(t|X_i, Z_i) = \lambda_0 \exp(Z_i^\top \beta) + X_i^\top \alpha.$$

where:

- Z_i are covariates with multiplicative effects,
- β are multiplicative coefficients,
- X_i are covariates with additive effects,
- α are additive coefficients

4 Target estimands and oracle variable importance

We considered two estimators of variable importance, targeting distinct estimands: exclusion importance and permutation importance.

4.1 Model-agnostic framework

Definition

We consider the model-agnostic variable importance measure proposed by Wolock *et al.*, based on a covariate exclusion principle (Wolock, Gilbert, Simon, & Carone, 2024a). For a given prediction a predefined time horizon t_0 and covariate (or covariate group) X_s , the VIM quantifies the loss in predictive performance induced by removing X_s from the prediction model. Using flexible survival prediction framework, we defined the non-parametric VIM measure for X_s as

$$\psi_s(P_0) = V(f_0) - V(f_{0,-s})$$

where $V(f_0)$ denotes the prediction model fitted using the complete set of covariates, and $V(f_{0,-s})$ the model fitted after excluding X_s .

Thus, $\psi_s(P_0)$ measures the incremental predictive contribution of X_s at prediction time t_0 , with larger values indicating greater importance.

Details on nuisance parameter and oracle prediction functions estimation

We estimated the conditional survival functions F_0 and G_0 using several modeling strategies depending on the simulation setting:

The conditional event and censoring survival functions, denoted respectively by F_0 and G_0 , were estimated using the learning strategies summarized in [Table 1](#).

Table 1: Learning strategies used to estimate nuisance parameters.

Method	R implementation	Reference
Aalen model	<code>timereg::aalen()</code>	(Therneau, 2007)
Cox-Aalen model	<code>pec::predictSurvProb()</code> <code>timereg::cox.aalen()</code>	(Therneau, 2007)
Survival Super Learner	<code>survivalSL::survivalSL</code> <code>survivalSL::predict.slttime()</code>	(Foucher & Sabathe, 2026)
Global survival stacking	<code>survML::stackG()</code>	(Wolock, Gilbert, Simon, & Carone, 2024b)

For time-horizon VIMs, the full oracle prediction function f_0 is a simple transformation of F_0 and was not estimated separately in the experiments. We estimated the residual oracle prediction function, $f_{0,s}$, using Super Learner regression ([Laan, Polley, & Hubbard, 2007](#)).

[Table 2](#) gives the algorithms included in the Super Learner library for global survival stacking and estimation of the residual oracle prediction function $f_{0,s}$ for VIM estimation. Five-fold cross-validation was used to determine the optimal convex combination of these learners that minimized cross-validated squared-error loss, as described in [Wolock et al. \(2024a\)](#).

Table 2: Algorithms used in the Super Learner for global survival stacking and for estimation of the residual oracle prediction function for VIM estimation.

Algorithme name	Algorithm description	Tuning parameters
<code>SL.mean</code>	Marginal mean	NA
<code>SL.gam</code>	Generalized additive model	NA
<code>SL.ranger</code>	Random forest	default
<code>SL.xgboost</code>	Gradient-boosted trees	<code>ntrees</code> $\in \{250, 500, 1000\}$ <code>max_depth</code> $\in \{1, 2\}$

[Table 3](#) gives the algorithms included in the survival Super Learner library ([Foucher & Sabathe, 2026](#)).

Table 3: Algorithms used in the survival Super Learner.

Algorithm name	Algorithm description	Tuning parameters
LIB_COXall	Usual Cox model	Default
LIB_PHexponential	Exponential-distributed PH model	Default
LIB_RSf	Survival random forest	Default

For global survival stacking, estimates for distributions were produced simultaneously. The Aalen and Cox-Aalen models, as well as the random survival forest algorithm were fitted twice, once to estimate F_0 and once to estimate G_0 .

We used the sample splitting procedure to compute VIM point and standard error estimates, from which we computed nominal 95% CI Wald-type confidence intervals.

4.2 Time horizon

Variable importance was evaluated at a fixed time horizon corresponding to the 75th percentile of the true event-time distribution, approximated using a Monte Carlo sample of size 2×10^7 . This choice focuses inference on a late-risk horizon while avoiding the numerical instability associated with the tail of the survival distribution. Restricting evaluation to a single horizon also substantially reduces computational burden.

4.3 Performance metrics

Variable importance was evaluated using the *Brier score*, $BS(t)$ to captures both discrimination and calibration. We additionally used the *cumulative/dynamic area under the ROC curve*, $AUC(t)$ to estimate variable importance measures, which evaluate discrimination independently of calibration.

The **Brier score** at time t is the expected squared difference between the observed survival status at time t and the predicted survival probability, with inverse probability of censoring weights to account for right censoring.

The **cumulative/dynamique AUC** at time t is defined as the probability that, among a randomly selected pair consisting of one individual who experiences the event before t and one who survives beyond t , the predicted risk is higher for the former.

4.4 Oracle variable importance

We generated a large Monte Carlo sample ($n = 2 \times 10^7$) from the true data-generating model to approximate population-level quantities. Using this sample, we computed the oracle predictive performance as well as the oracle VIM measure for each variables.

5 Performance evaluation

The performance of the proposed methods was evaluated across $B = 1000$ Monte Carlo replications using the metrics summarized in Table 4. These criteria assess estimation accuracy, inferential validity, ranking performance, and computational efficiency.

Table 4: Performance metrics used to evaluate estimation accuracy, inference, ranking ability, and computational efficiency across simulation replicates.

Metric	Definition	Formula
Bias	Average deviation between the estimator and the true parameter	$\frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \theta)$
RMSE	Root mean squared error measuring overall estimation accuracy	$\sqrt{\frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \theta)^2}$
Coverage	Proportion of confidence intervals containing the true parameter	$\frac{1}{B} \sum_{b=1}^B \mathbf{1}(\theta \in CI^{(b)})$
CI width	Average length of confidence intervals	$\frac{1}{B} \sum_{b=1}^B (U^{(b)} - L^{(b)})$
Type I Error	Probability of rejecting H_0 when it is true	$\frac{1}{B} \sum_{b=1}^B \mathbf{1}(p^{(b)} < \alpha)$ under H_0
Power	Probability of rejecting H_0 when the alternative is true	$\frac{1}{B} \sum_{b=1}^B \mathbf{1}(p^{(b)} < \alpha)$ under H_1
Rank correlation	Correlation between true and estimated parameter rankings	
Runtime	Average computational time per replication	$\frac{1}{B} \sum_{b=1}^B t^{(b)}$

B denotes the number of Monte Carlo replications, θ the true parameter value, $\hat{\theta}^{(b)}$ the estimate obtained at replication b , $CI^{(b)} = [L^{(b)}, U^{(b)}]$ the corresponding confidence interval, and α the nominal significance level.

6 Simulation scenarios

The main scenarios of the simulation study are summarized in [Table 5](#).

Table 5: Summary of the main scenarios

	Effect type	Covariates	Coefficient	Nuisance parameters estimation
S1	Additive hazards (Aalen model)	$X_1, X_2, X_3, X_4 \sim \mathcal{B}(0.4)$	$\alpha_1 = 0.1$ (strong effect) $\alpha_2 = 0.05$ (moderate effect) $\alpha_3 = 0.005$ (weak effect) $\alpha_4 = 0$ (null effect)	Aalen model Survival Super Learner Global survival stacking
S2	Mixed additive and multiplicative hazards (Cox-Aalen model)	$X_1, X_2, X_3 \sim \mathcal{B}(0.4)$ $Z_1, Z_2 \sim \mathcal{B}(0.4)$ $Z_3 \sim \mathcal{N}(0, 1)$	$\alpha_1 = 0.1$ (strong effect) $\alpha_2 = 0.05$ (moderate effect) $\alpha_3 = 0$ (null effect) $\beta_1 = 1$ (strong effect) $\beta_2 = 0.5$ (moderate effect) $\beta_3 = 0$ (null effect)	Cox-Aalen model Survival Super Learner Global survival stacking

6.1 Scenario 1 (S1) - Additive hazards

This scenario evaluates the robustness of Wolock’s model-agnostic variable importance measure (VIM) under correct model specification. Specifically, we assess whether competing nuisance-function estimators recover the true exclusion VIM when the data-generating mechanism follows an **additive hazards model**.

Four binary covariates were generated *independently* as follows: $X_1, X_2, X_3 \sim \mathcal{B}(0.4)$.

Events times were simulated under the following time-constant additive hazards model:

$$\lambda(t|x) = \lambda_0(t) + \alpha_1(t)X_1 + \alpha_2(t)X_2 + \alpha_3(t)X_3 + \alpha_4(t)X_4$$

with $\lambda_0(t) = 0.05$ (constant) and $\alpha_j(t) = (0.1, 0.05, 0.005, 0)$.

Independent censoring times were generated as $C \sim \text{Uniform}(0, c_{max})$. The parameter c_{max} was calibrated using a large Monte Carlo sample from the data-generating mechanism to achieve an expected censoring proportion of approximately 20%, and the calibrated value was then fixed across all simulation replicates.

We compared several approaches for estimating the nuisance functions required for VIM computation, including the Aalen additive hazards model, the survival Super Learner, and global survival stacking.

The approximate true values of variable importance based on $AUC(t)$ and, Brier score ($BS(t)$) under this scenario are provided in [Table 6](#).

Table 6: Approximate values of variable importance for S1. $\tau=14.97$.

VIM type	Variable	$\tilde{V}^{MC}(f_0)$	$\tilde{V}^{MC}(f_{0,-s})$	$\tilde{\psi}_{0,s}^{MC}$
BS(t)	X_1	-0.14	-0.18	0.041
AUC(t)	X_1	0.73	0.60	0.121
BS(t)	X_2	-0.14	-0.16	0.021
AUC(t)	X_2	0.73	0.67	0.053
BS(t)	X_3	-0.14	-0.14	0.002
AUC(t)	X_3	0.73	0.72	0.003
BS(t)	X_4	-0.14	-0.14	0.000
AUC(t)	X_4	0.73	0.73	0.000

6.2 Scenario 2 (S2) - Mixed additive and multiplicative effects

This scenario evaluate the sensitivity of the exclusion VIM when the true hazard follows a **Cox-Aalen model**.

The covariates are generated *independently* as follows:

- $X_1, X_2, X_3, Z_1, Z_2 \sim \mathcal{B}(0.4)$
- $Z_3 \sim \mathcal{N}(0, 1)$

To generate mixed additive and multiplicative effects, event times were simulated under a Cox-Aalen model with constant baseline hazard: $\lambda_0(t) = 0.05$. Additive effects were assigned to X_1 , X_2 and X_3 with coefficients $\alpha_1(t) = 0.1$, $\alpha_2 = 0.05$ and $\alpha_3(t) = 0$, respectively. Multiplicative (proportional hazards) effects were assigned to Z_1 , Z_2 and Z_3 with coefficients $\beta_1 = 1$, $\beta_2 = 0.5$ and $\beta_3 = 0$.

Independent censoring times were generated as $C \sim \text{Uniform}(0, c_{max})$. The parameter c_{max} was calibrated using a large Monte Carlo sample from the data-generating mechanism to achieve an expected censoring proportion of approximately 20%, and the calibrated value was then fixed across all simulation replicates.

We compared several approaches for estimating the nuisance functions required for VIM computation, including the Cox-Aalen model, the survival Super Learner, and global survival stacking.

The approximate true values of variable importance based on $AUC(t)$ and, Brier score ($BS(t)$) under this scenario are provided in [Table 7](#).

Table 7: Approximate values of variable importance for S1. $\tau=9.10$.

VIM type	Variable	$\tilde{V}^{MC}(f_0)$	$\tilde{V}^{MC}(f_{0,-s})$	$\tilde{\psi}_{0,s}^{MC}$
BS(t)	X_1	-0.13	-0.17	0.037
AUC(t)	X_1	0.78	0.72	0.066
BS(t)	X_2	-0.13	-0.15	0.019
AUC(t)	X_2	0.78	0.76	0.027
BS(t)	X_3	-0.13	-0.13	0.000
AUC(t)	X_3	0.78	0.78	0.000
BS(t)	Z_1	-0.13	-0.17	0.036
AUC(t)	Z_1	0.78	0.72	0.065
BS(t)	Z_2	-0.13	-0.15	0.013
AUC(t)	Z_2	0.78	0.76	0.019
BS(t)	Z_3	-0.13	-0.13	0.000
AUC(t)	Z_3	0.78	0.78	0.000

6.3 Additional numerical experiments

6.3.1 Sensitivity to low-prevalence covariates

This scenario evaluates the overall performance of the VIM procedure to ***low-prevalence covariates*** with ***heterogeneous prognostic effects***, ranging from null to strong, under both additive and mixed additive-multiplicative hazard structures. The objective is to assess whether the model-agnostic VIM approach can identify rare yet highly prognostic covariates, discriminate between moderate and strong effects, and correctly detect non-informative variables.

Covariates were generated with different marginal prevalences and effect sizes, leading to the simulation settings summarized in [Table 8](#). These experiments were designed to assess the sensitivity of the proposed approach to low-prevalence covariates and varying levels of prognostic signal.

Table 8: Simulation settings assessing sensitivity to low-prevalence covariates

Prevalence category	π_j	Effect category	α	β
Very rare	$\pi = 0.01$	Strong	0.10	1.0
Very rare	$\pi = 0.01$	Moderate	0.05	0.5
Very rare	$\pi = 0.01$	Null	0.00	0.0

Table 8: Simulation settings assessing sensitivity to low-prevalence covariates

Prevalence category	π_j	Effect category	α	β
Rare	$\pi = 0.10$	Strong	0.10	1.0
Rare	$\pi = 0.10$	Moderate	0.05	0.5
Rare	$\pi = 0.10$	Null	0.00	0.0

6.3.2 Sensitivity to covariate correlations

For each scenario, we additionally evaluated the sensitivity of the VIM to correlations between covariates. Correlations were introduced between selected covariates while preserving their marginal distributions (binary or continuous). Specifically, we considered both weak and moderate correlation dependence between predictors, with pairwise correlations set to $\rho = 0.3$ and $\rho = 0.6$, respectively. This analysis allows us to assess the robustness of the estimated VIM in the presence of correlated covariates.

References

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